

LR	Locked rotor
L1, L2, L3	Line conductors of a three-phase AC system
M	Asynchronous motor or group of asynchronous motors
MO	Without motor
MV	Medium-voltage, medium-voltage side of a transformer
N	Neutral of a three-phase AC system, star point of a generator or a transformer
P	Generator with compound excitation
PF	Power station unit with full size converter
PFO	Without current sources of power station units with full size converters
PV	Photovoltaic power station unit
Q	Feeder connection point
R	Short-circuit limiting reactor
S	Power station unit (generator and unit transformer with on-load tap-changer)
SO	Power station unit (generator and unit transformer with constant transformation ratio or off-load taps)
T	Transformer
WA	Wind power station unit with asynchronous generator
WD	Wind power station unit with doubly fed asynchronous generator
WF	Wind power station unit with full size converter

4.4 Superscripts

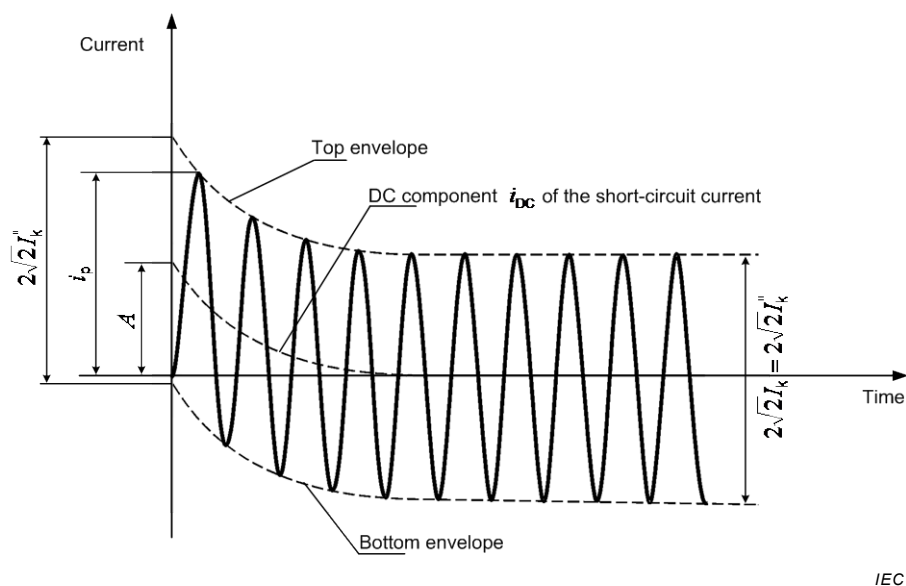
"	Subtransient (initial) value
'	Resistance or reactance per unit length
b	Before the short circuit

5 Characteristics of short-circuit currents: calculating method

5.1 General

A complete calculation of short-circuit currents should give the currents as a function of time at the short-circuit location from the initiation of the short circuit up to its end (see Figures 1 and 2), corresponding to the instantaneous value of the voltage before the short circuit.

NOTE In real networks the short-circuit current can deviate from the current shape of Figures 1 and 2.



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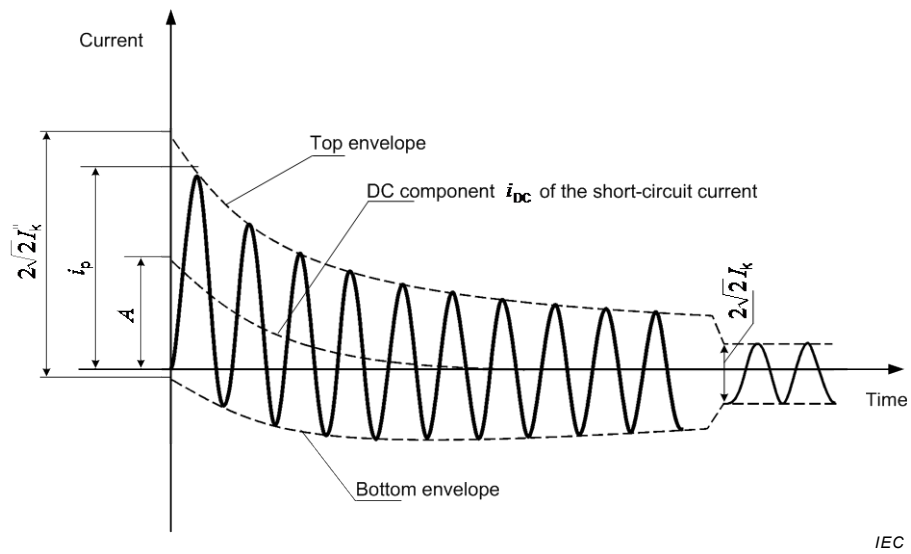
Key

- I_k'' initial symmetrical short-circuit current
- i_p peak short-circuit current
- I_k steady-state short-circuit current
- i_{DC} DC component of short-circuit current
- A initial value of the DC component i_{DC}

Figure 1 – Short-circuit current of a far-from-generator short circuit with constant AC component (schematic diagram)

In most practical cases a determination like this is not necessary. Depending on the application of the results, it is of interest to know the rms value of the symmetrical AC component and the peak value i_p of the short-circuit current following the occurrence of a short circuit. The highest value i_p depends on the time constant of the decaying aperiodic component and the frequency f , that is on the ratio R/X or X/R of the short-circuit impedance Z_k , and is reached if the short circuit starts at zero voltage. i_p also depends on the decay of the symmetrical AC component of the short-circuit current.

In meshed networks there are several direct-current time constants. That is why it is not possible to give an easy method of calculating i_p and i_{DC} . Special methods to calculate i_p with sufficient accuracy are given in Clause 8.

**Key**

- I_k'' initial symmetrical short-circuit current
- i_p peak short-circuit current
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- A initial value of the DC component i_{DC}

Figure 2 – Short-circuit current of a near-to-generator short-circuit with decaying AC component (schematic diagram)

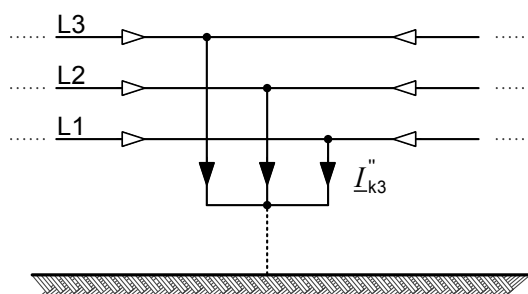


Figure 3a – Three-phase short circuit

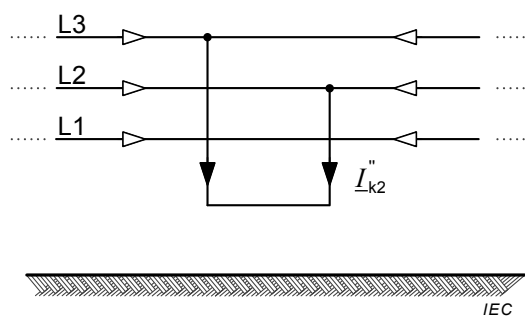


Figure 3b – Line-to-line short circuit

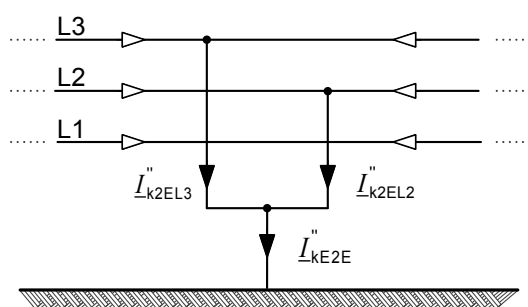


Figure 3c – Line-to-line short circuit with earth connection

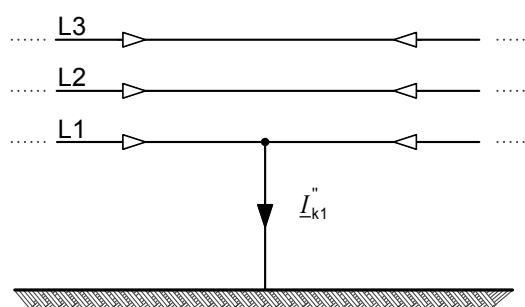


Figure 3d – Line-to-earth short circuit

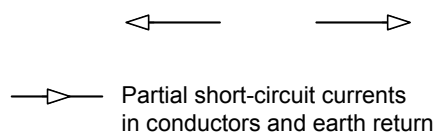
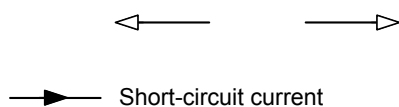


Figure 3 – Characterization of short-circuits and their currents

5.2 Calculation assumptions

The calculation of maximum and minimum short-circuit currents is based on the following simplifications.

- For the duration of the short circuit there is no change in the type of short circuit involved, that is, a three-phase short-circuit remains three-phase and a line-to-earth short circuit remains line-to-earth during the time of short circuit.
- For the duration of the short circuit, there is no change in the network involved.
- The impedance of the transformers is referred to the tap-changer in main position.
- Arc resistances are not taken into account.
- Shunt admittances of non-rotating loads shall be neglected in the positive-, the negative- and the zero-sequence system.
- Line capacitances shall be neglected in the positive- and negative-sequence system. Line capacitances in the zero-sequence system shall be taken into account in low-impedance earthed networks having an earth-fault factor (see IEC 60027-1) higher than 1,4.
- Magnetising admittances of transformers shall be neglected in the positive- and negative-sequence system.

Despite these assumptions being not strictly true for the power systems considered, the result of the calculation does fulfil the objective to give results which are generally of acceptable accuracy.

For balanced and unbalanced short-circuits as shown in Figure 3, the short-circuit currents can be calculated by the application of symmetrical components (see 5.3.2).

The impedances of the equipment in superimposed or subordinated networks are to be divided or multiplied by the square of the rated transformation ratio t_r . Voltages and currents are to be converted only by the rated transformation ratio t_r . If there are several transformers with slightly differing rated transformation ratios (t_{rT1} , t_{rT2} , ... t_{rTn}), in between two systems, the arithmetic mean value can be used.

For per unit or other similar unit systems, no transformation is necessary if these systems are coherent, i.e. $U_{rTHV}/U_{rTLV} = U_{nHV}/U_{nLV}$ for each transformer in the system with partial short-circuit currents. U_{rTHV}/U_{rTLV} is normally not equal to U_{nHV}/U_{nLV} (see IEC TR 60909-2 and the examples given in IEC TR 60909-4).

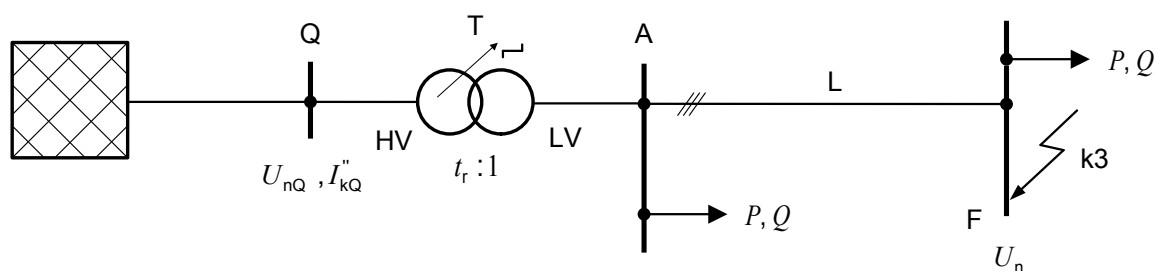
5.3 Method of calculation

5.3.1 Equivalent voltage source at the short-circuit location

The method used for calculation is based on the introduction of an equivalent voltage source at the short-circuit location. The equivalent voltage source is the only active voltage of the system. All network feeders, synchronous and asynchronous machines are replaced by their internal impedances (see Clause 6).

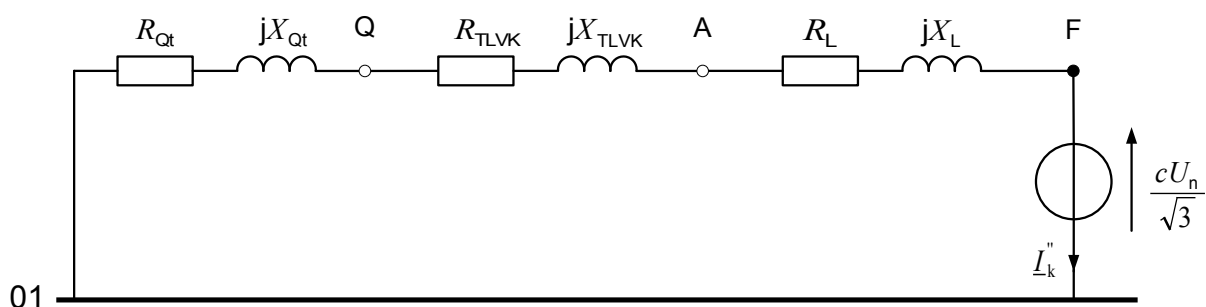
In all cases it is possible to determine the short-circuit current at the short-circuit location F with the help of an equivalent voltage source. Operational data and the load of consumers, tap-changer position of transformers, excitation of generators, and so on, are dispensable; additional calculations about all the different possible load flows at the moment of short-circuit are superfluous.

Figure 4 shows an example of the equivalent voltage source at the short-circuit location F as the only active voltage of the system fed by a transformer without or with on-load tap-changer. All other active voltages in the system are short-circuited. Thus the network feeder in Figure 4a is represented by its internal impedance Z_{Qt} transferred to the LV-side of the transformer (see 6.2) and the transformer by its impedance referred to the LV-side (see 6.3). The shunt admittances of the line, of the transformer and of the non-rotating loads are not considered in accordance with 5.2 e) to g).



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Figure 4a – System diagram



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Figure 4b – Equivalent circuit diagram of the positive-sequence system

NOTE The index (1) for the impedances of the positive-sequence system is omitted. 01 marks the positive-sequence neutral reference. The impedances of the network feeder and the transformer are converted to the LV-side and the last one is also corrected with K_T (see 6.3.3).

Figure 4 – Illustration for calculating the initial symmetrical short-circuit current I_k'' in compliance with the procedure for the equivalent voltage source

If there are no national standards, it seems adequate to choose a voltage factor c according to Table 1, considering that the highest voltage in a normal (undisturbed) system does not differ, on average, by more than approximately +5 % (some LV systems) or +10 % (some HV systems) from the nominal system voltage U_n .

Table 1 – Voltage factor c

Nominal system voltage U_n	Voltage factor c for the calculation of	
	maximum short-circuit currents $c_{\max}^a$	minimum short-circuit currents $c_{\min}$
Low voltage 100 V to 1 000 V (IEC 60038:2009, Table 1)	1,05 ^c 1,10 ^d	0,95 ^c 0,90 ^d
High voltage ^b >1 kV to 230 kV (IEC 60038:2009, Tables 3, 4)	1,10	1,00
High voltage ^{b, e} > 230 kV (IEC 60038:2009, Table 5)	1,10	1,00
^a $c_{\max} U_n$ should not exceed the highest voltage U_m for equipment of power systems. ^b If no nominal system voltage is defined $c_{\max} U_n = U_m$ or $c_{\min} U_n = 0,90 \cdot U_m$ should be applied. ^c For low-voltage systems with a tolerance of $\pm 6\%$, for example systems renamed from 380 V to 400 V. ^d For low-voltage systems with a tolerance of $\pm 10\%$. ^e For nominal system voltages related to $U_m > 420$ kV, the voltage factors c are not defined in this standard.		

5.3.2 Symmetrical components

In three-phase AC systems the calculation of the current values resulting from balanced and unbalanced short circuits is simplified by the use of symmetrical components. This postulates that the electrical equipment has a balanced structure, for example in the case of transposed overhead lines. The results of the short-circuit current calculation have an acceptable accuracy also in the case of untransposed overhead lines.

Using this method, the currents in each line conductor are found by superposing the currents of the three symmetrical component systems:

- positive-sequence current $\underline{I}_{(1)}$;
- negative-sequence current $\underline{I}_{(2)}$;
- zero-sequence current $\underline{I}_{(0)}$.

Taking the line conductor L1 as reference, the currents $\underline{I}_{L1}$, $\underline{I}_{L2}$, and $\underline{I}_{L3}$ are given by

$$\underline{I}_{L1} = \underline{I}_{(1)} + \underline{I}_{(2)} + \underline{I}_{(0)} \quad (1)$$

$$\underline{I}_{L2} = \underline{a}^2 \cdot \underline{I}_{(1)} + \underline{a} \cdot \underline{I}_{(2)} + \underline{I}_{(0)} \quad (2)$$

$$\underline{I}_{L3} = \underline{a} \cdot \underline{I}_{(1)} + \underline{a}^2 \cdot \underline{I}_{(2)} + \underline{I}_{(0)} \quad (3)$$

with $\underline{a} = -\frac{1}{2} + j\frac{1}{2}\sqrt{3}$ $\underline{a}^2 = -\frac{1}{2} - j\frac{1}{2}\sqrt{3}$

Each of the three symmetrical component systems has its own impedance.

The following types of unbalanced short-circuits are treated in this standard: